

Understanding My Affective Ising Model Project

A personal guide from zero - simple English

Course: Stochastic Modelling

Paper: Loossens et al. (2020), The Affective Ising Model

Status: energy, gradient and deterministic drift completed

Purpose

This is not text for submission. It is a study document that explains what the paper does, what my project tries to achieve, the course theory behind it, my current results, and all remaining work.

Updated 24 July 2026

1. My project in one page

The human explanation

Imagine a ball moving on a landscape. Valleys are stable emotional states. Hills separate those states. The landscape pulls the ball downhill, while random noise shakes it. A strong random movement or an external push can move the ball from one valley to another.

Memory map

Ball = affective state; landscape = free energy; slope = gradient; downhill pull = drift; random shaking = diffusion; valley = homebase; crossing = switching.

My objective

First I reproduce the mathematical and stochastic mechanism of the Affective Ising Model (AIM). Then I add a temporary negative external input and study whether it changes switching and recovery.

Research question

Can a temporary negative input increase the probability of moving from a high-positive-affect homebase to a high-negative-affect homebase, and can it delay recovery?

Reproduction versus original contribution

Reproduction	My new contribution
Free energy and gradient	Temporary negative input
Stationary theoretical density	Control, weak pulse, strong pulse
Euler-Maruyama trajectories	Switching probability
Theory-simulation validation	Recovery time; optional temporal patterns

Current position

The static foundation is complete. The stationary density, stochastic simulation and external-input experiment are still unfinished. Current overall progress is approximately 25%.

2. Stochastic modelling theory from zero

Deterministic model

A deterministic model gives the same trajectory every time when the initial condition is the same.

$$dx/dt = f(x, t)$$

Stochastic model

A stochastic model includes random variation. Two runs can start at the same point and then follow different paths.

$$dX_t = a(X_t, t) dt + b(X_t, t) dW_t$$

- **a(X,t)**: drift, the systematic direction.
- **b(X,t)**: diffusion amplitude, the strength of randomness.
- **W(t)**: Wiener process or Brownian motion.

Trajectory versus probability

One simulation is one trajectory or realization. A probability density describes the possible states across many repetitions. A beautiful single trajectory is not enough evidence for a stochastic result.

Wiener increment

$$\begin{aligned} \Delta W &\sim \text{Normal}(0, \Delta t) \\ \Delta W &= \sqrt{\Delta t} * G, \text{ with } G \sim \text{Normal}(0,1) \end{aligned}$$

The deterministic part scales with Δt . The random part scales with $\sqrt{\Delta t}$. This square-root rule is essential in stochastic simulation.

Course connection

The Langevin/SDE description follows individual random paths. The Fokker-Planck equation follows the probability density of the same system.

3. Potential landscapes, equilibrium and Euler-Maruyama

Potential and gradient

The gradient points toward the fastest increase of a potential. Its negative points downhill.

```
gradient U = (partial U/partial x1, partial U/partial x2)
downhill direction = -gradient U
```

Stable and unstable points

- A local minimum is stable: nearby drift points toward it.
- A maximum is unstable: nearby drift points away.
- A saddle or barrier separates attraction basins.

Stationary density

A stationary density no longer changes with time. Individual trajectories still move, but the overall probability pattern stays constant.

```
p(y1,y2) = exp[-beta F(y1,y2)] / Z
```

- Low energy gives high probability.
- High energy gives low probability.
- Z normalizes the total probability to one.
- Two energy minima can generate two probability peaks.

Euler-Maruyama

```
X[n+1] = X[n] + a(X[n]) Delta t
          + b(X[n]) sqrt(Delta t) G[n]
```

This is the stochastic version of a simple Euler step. Later I will apply it to both PA and NA.

Main validation principle

A long correct simulation should visit the same regions predicted by the theoretical stationary density.

4. What the AIM paper is about

Problem

The paper studies how positive affect (PA) and negative affect (NA) change over time. Real affect data can be skewed, V-shaped or multimodal. Simple Gaussian models cannot naturally represent all these patterns.

Model idea

- One abstract pool represents positive affect.
- One abstract pool represents negative affect.
- Units excite other units in the same pool.
- The two pools inhibit one another.
- External events can affect the pools.

The “binary neurons” are an abstract computational description. They are not a literal biological map.

Data and fitting

The authors fitted the model to many experience-sampling time series. Participants repeatedly reported emotions during daily life. The fitting used numerical likelihood calculations and Differential Evolution, with many repeated optimizations.

Main results

- The AIM can represent Gaussian-like, skewed, V-shaped and bimodal distributions.
- It reproduced nonlinear empirical features better than Gaussian benchmark models.
- It had better predictive performance for a majority of the analysed time series.
- It provides both equilibrium probability and continuous-time dynamics.

Why my extension fits the paper

The paper defines time-dependent external inputs $B1(t)$ and $B2(t)$, but does not analyse known events because the datasets do not contain enough objective event information.

Justification

My project develops a mechanism already present in the AIM instead of adding an unrelated modification.

5. AIM equations and parameters

State variables

$0 \leq y_1 \leq 1$ positive affect (PA)
 $0 \leq y_2 \leq 1$ negative affect (NA)

Free energy

$$F = -L_1 y_1^2 + T_1 y_1 - L_2 y_2^2 + T_2 y_2 + L_{12} y_1 y_2 + (N_1/\beta)h(y_1) + (N_2/\beta)h(y_2)$$

$$h(y) = y \log(y) + (1-y) \log(1-y)$$

Parameter	Simple meaning
L1, L2	Self-excitation inside PA and NA.
L12	Mutual inhibition between PA and NA.
T1, T2	Activation thresholds.
N1, N2	Effective pool sizes.
beta	Inverse-temperature scale; fixed to 1 in the paper.

Why the logarithms matter

The entropy terms come from large pools of binary units. They make the landscape nonlinear. They are not finite exactly at 0 or 1, so the current numerical grids use values such as 0.01 to 0.99.

Gradient

$$dF/dy_1 = -2 L_1 y_1 + T_1 + L_{12} y_2 + (N_1/\beta) \log[y_1/(1-y_1)]$$

$$dF/dy_2 = -2 L_2 y_2 + T_2 + L_{12} y_1 + (N_2/\beta) \log[y_2/(1-y_2)]$$

The gradient points uphill. The deterministic drift uses the downhill direction.

Current synthetic parameters

$L_1=L_2=300, L_{12}=100, T_1=T_2=250, N_1=N_2=100, \beta=1$

Correct interpretation

The paper defines these parameter types, but it does not publish the exact participant vectors used for its example figures. My values are a transparent synthetic symmetric benchmark, not estimates for a real person.

6. What I have programmed and validated

Code organisation

```
notebooks/01_free_energy_landscape.py
src/aim/model.py
results/figures/free_energy_landscape.png
results/figures/drift_field.png
```

Energy calculation

I evaluated F on a two-dimensional PA-NA grid. I first used nested loops and then NumPy meshgrid/vectorization. Both methods agreed, while the vectorized version was much faster.

Gradient validation

I checked the analytical gradient using central finite differences at $(0.3, 0.7)$, with $h=10^{-6}$.

```
Numerical dF/dy1 = 55.2702139629
Analytical dF/dy1 = 55.2702139613
```

```
Numerical dF/dy2 = -55.2702139522
Analytical dF/dy2 = -55.2702139613
```

The differences are approximately 10^{-9} and both tests pass. This is strong evidence that the analytical derivatives and signs are correct.

Deterministic drift

I evaluated $-\text{gradient } F$ on a smaller grid and plotted arrows. I normalized the arrows so that direction is visible; therefore their lengths should not be interpreted as drift strength.

What is validated now

The static energy function, analytical gradient and downhill direction are internally consistent.

7. Current result: free-energy landscape

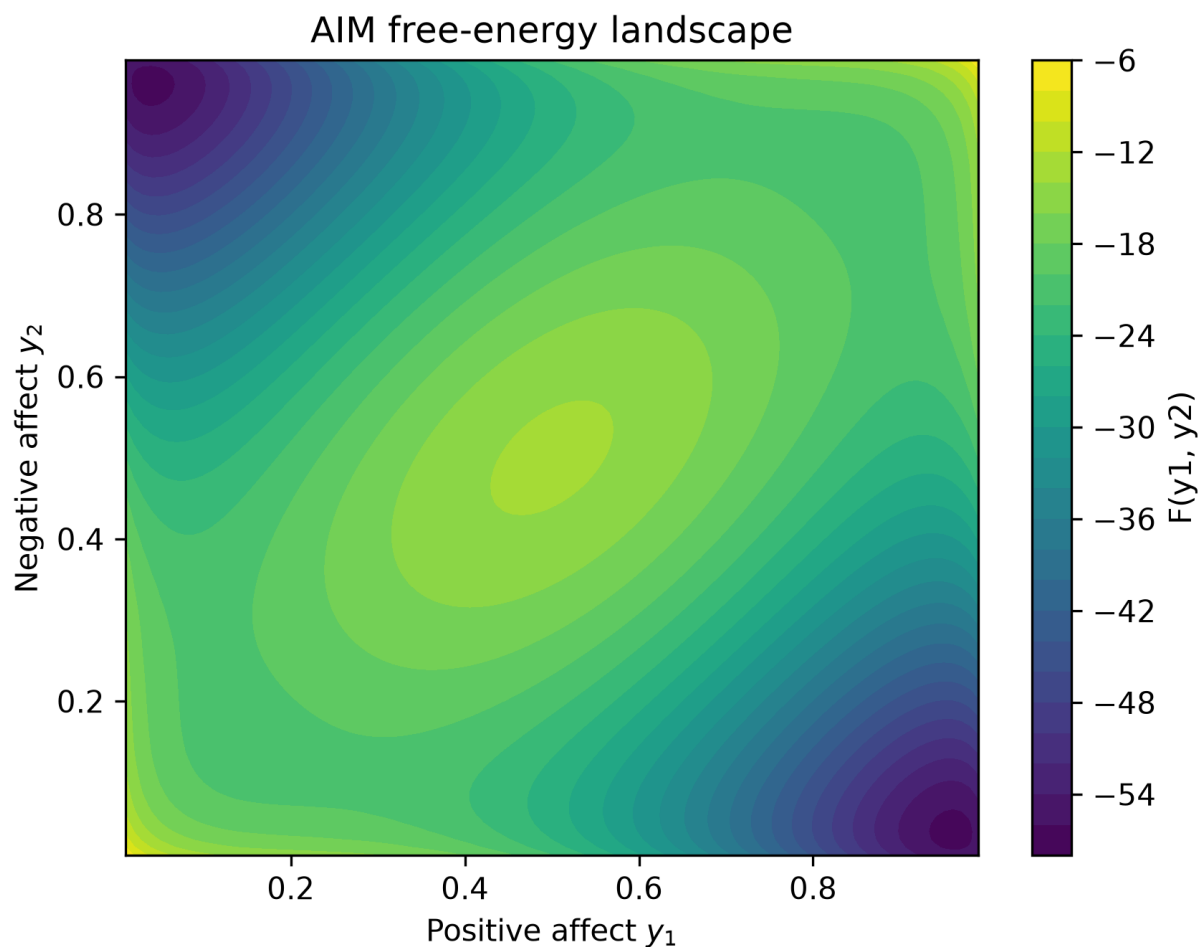


Figure 1. Synthetic symmetric AIM free-energy landscape.

What I see

- Two dark low-energy regions near opposite corners.
- A higher-energy central region.
- One minimum represents high PA/low NA.
- The other represents low PA/high NA.

Interpretation

The selected configuration is bistable. The central high-energy region acts as a barrier between two homebases.

Not proved yet

This figure does not show random switching, recovery or the effect of a negative input. Those require stochastic simulation.

8. Current result: deterministic drift field

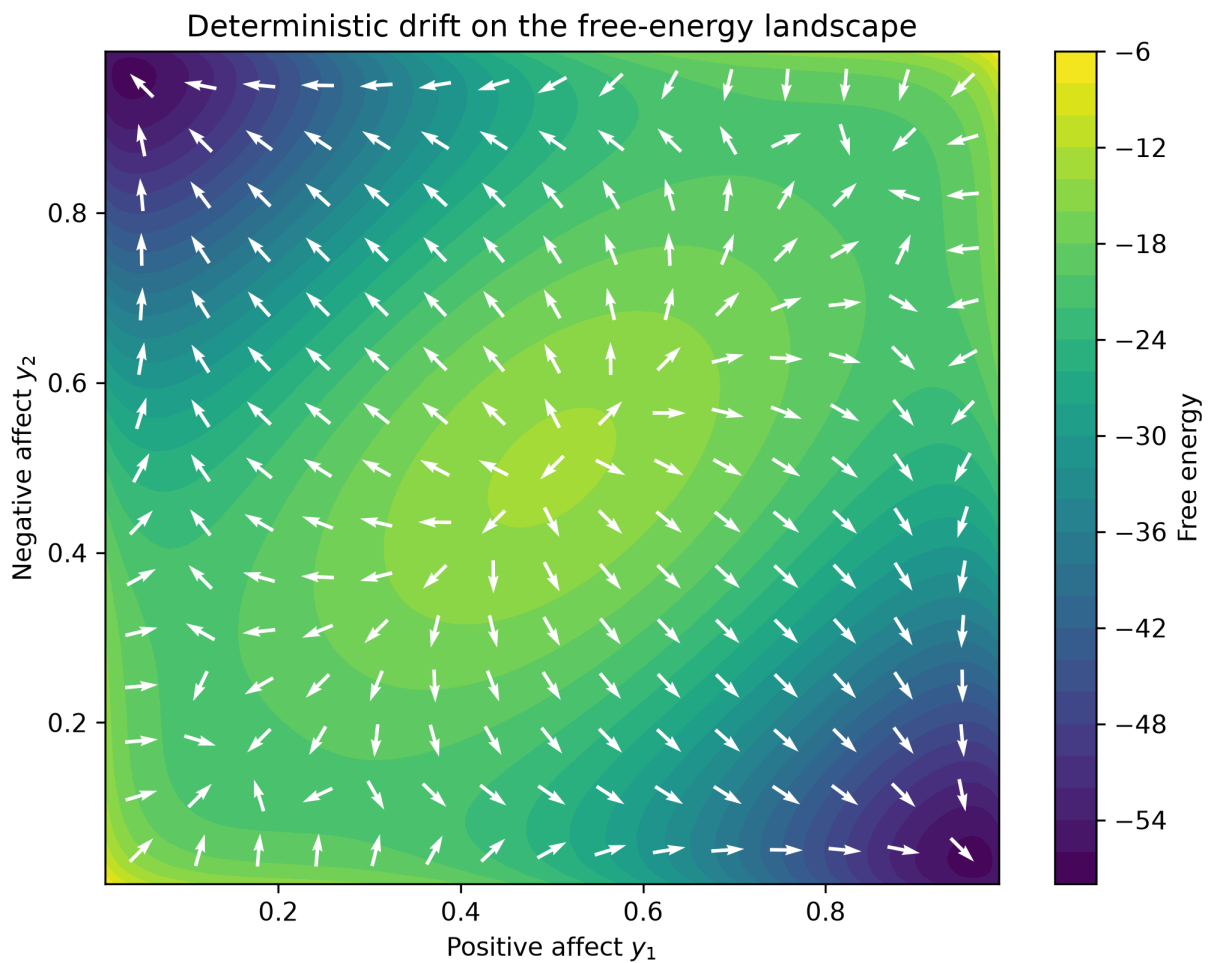


Figure 2. Normalized negative-gradient directions.

What I see

- Arrows move away from the high-energy centre.
- Arrows converge toward both low-energy regions.
- The field is consistent with downhill potential motion.
- Two attraction basins are visible.

Why the arrows have equal size

Every vector was divided by its own magnitude. This keeps the direction but removes information about drift strength.

Current conclusion

The gradient test and drift figure validate the static mathematical implementation before adding noise.

9. What remains and the order of work

Stage	Status	Output
Energy and gradient	Complete	Landscape, test, drift
Stationary density	In progress	stationary_density.png
Euler-Maruyama	Not started	PA/NA trajectories
Theory vs simulation	Not started	Density comparison
External input	Not started	Switching/recovery
Presentation	Not started	Final slides

Step 1: stationary density

- Calculate F on a grid.
- Calculate $\exp(-\beta F)$ safely.
- Normalize using grid-cell area.
- Check that the numerical integral is approximately 1.
- Plot two expected probability peaks.

Step 2: stochastic simulation

- Implement the exact AIM SDE.
- Use Euler-Maruyama.
- Keep y_1 and y_2 inside $[0,1]$.
- Test a smaller time step.

Step 3: validation

- Run a long trajectory.
- Create a two-dimensional empirical histogram.
- Compare empirical and theoretical densities.

Step 4: new experiment

- Start near the positive homebase.
- Apply a temporary negative input.
- Compare control, weak pulse and strong pulse.
- Repeat with many random seeds.

10. Planned experiment and measurements

Condition	Input	Purpose
Control	No external input	Baseline random switching
Weak pulse	Small temporary B2(t)	Mild perturbation
Strong pulse	Larger temporary B2(t)	Test increased switching

Why many trajectories?

One trajectory can switch by chance. Probabilities require many independent simulations.

Main measurement

$$\text{switching probability} = \text{number of switching trajectories} / \text{total trajectories}$$

Additional measurements

- Persistent trapping: still in the negative basin at the end.
- Recovery time: time after input removal until returning.
- Residence time: continuous time inside the negative basin.

Optional extension

If the minimal experiment is finished, compare a strong short pulse with a weaker long pulse while keeping total input approximately equal.

Do not decide the conclusion early

The simulation must determine the result. I should not assume that one input pattern is worse before running the experiment.

11. Claims, limitations and defence questions

What I can claim now

- The free-energy equation is implemented.
- The gradient agrees with finite differences.
- The synthetic configuration creates two minima.
- The negative gradient points toward those minima.

What I cannot claim yet

- Negative input causes switching.
- Strong input delays recovery.
- The stochastic simulation matches theory.
- The parameters describe a real person.
- Model time equals hours or days.

Limitations

- Synthetic parameters.
- Published example vectors unavailable.
- Possible boundary effects.
- Euler-Maruyama discretization error.
- Abstract model and cautious psychological interpretation.

Questions I must answer

Question	Short answer
Why stochastic?	The AIM combines deterministic drift with Wiener-process noise.
What is a homebase?	A stable, frequently visited region; a local energy minimum.
Why -gradient F?	The gradient points uphill, so its negative points downhill.
Why finite differences?	They independently detect derivative or sign errors.
Are parameters empirical?	No. They are a documented synthetic benchmark.
Why compare density and simulation?	Agreement validates dynamics, noise and numerics.
What is new?	Controlled negative inputs and switching/recovery analysis.

12. Glossary and final checklist

Term	Meaning
Affect state	One point (y_1, y_2) .
Attractor/homebase	Stable region approached by nearby states.
Bistability	Two stable regions.
Drift	Systematic component of an SDE.
Diffusion	Random spreading component.
Wiener process	Continuous process with Gaussian increments.
Stationary density	Long-time probability pattern.
Euler-Maruyama	Numerical method for Ito SDEs.
Monte Carlo	Repeated random simulation.
Trajectory	One random realization over time.

Immediate checklist

- Finish 02_stationary_density.py.
- Verify total probability approximately 1.
- Save stationary_density.png.
- Implement and test Euler-Maruyama.
- Compare theoretical and simulated density.
- Run the negative-input experiment.
- Update this guide with real results.
- Prepare final figures and presentation.

Final memory sentence

The AIM is a stochastic dynamical system in which the affective state is pulled downhill by a free-energy landscape and randomly moved by diffusion. My project validates this mechanism and then tests temporary negative perturbations.

Sources

Main paper

Loossens, T., Mestdagh, M., Dejonckheere, E., Kuppens, P., Tuerlinckx, F., & Verdonck, S. (2020). The Affective Ising Model: A computational account of human affect dynamics. PLOS Computational Biology, 16(5), e1007860.

Course material

Stochastic Modelling course notes: Brownian motion, Langevin equations, Ito SDEs, drift and diffusion, Fokker-Planck equations, potential landscapes, stationary states, multidimensional systems and Euler-Maruyama.

Project material

Current Python code and generated figures in affective-ising-temporal-inputs.

Reminder

This document describes the project on 24 July 2026. Future simulation claims must be added only after the corresponding numerical results exist.